

A passive cell body plays an active role in microalgal swimming via non-reciprocal interactions

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The cell body of flagellated microalgae is commonly considered to act merely as a passive load during swimming, and a larger body size would simply reduce the speed. In this work, we use numerical simulations based on a boundary element method to investigate the effect of body–flagella hydrodynamic interactions (HIs) on the swimming performance of the biflagellate *Chlamydomonas reinhardtii*. We find that body–flagella HIs significantly enhance swimming speed and efficiency. As body size increases, the competition between the enhanced HIs and the increased viscous drag leads to an optimal body size for swimming. Based on the simplified three-sphere model, we further demonstrate that the enhancement by body–flagella HIs arises from an effective non-reciprocity: the body affects the flagella more strongly during the power stroke, while the flagella affect the body more strongly during the recovery stroke. Our results have implications for both microalgal swimming and laboratory designs of biohybrid microrobots.

Key words: swimming/flying, boundary integral methods, micro-organism dynamics

1. Introduction

The unicellular green alga *Chlamydomonas reinhardtii* has been widely used as a model organism for studying various problems in fluid dynamics and cell biology (Goldstein 2015), such as tactic behaviours (Leptos *et al.* 2023), flagellar dynamics (Quaranta, Aubin-Tam & Tam 2015; Sartori *et al.* 2016) and single-cell flow fields (Drescher *et al.* 2010; Wei *et al.* 2019, 2021). The wild-type cell of *C. reinhardtii* has two flagella anchored

at the anterior side of the spheroidal cell body. During swimming, the two flagella beat approximately in a two-dimensional plane, with a breaststroke-like gait consisting of distinct power and recovery strokes. It is now well established that flagellar beating is driven by distributed molecular motors, a characteristic conserved across eukaryotic cells. Studies of the swimming mechanisms of green algae have inspired numerous designs of artificial flagella and microswimmers (see e.g. Diaz *et al.* 2021; Moreau *et al.* 2024).

The analysis of eukaryotic flagellar dynamics often assumes a fixed flagellar base in space. Machin (1958) solved the bending waves of an elastic beam actuated at the boundary. Although this boundary-driven assumption is a significant simplification of realistic flagellar dynamics, later developments reveal the effects of various physical elements on flagellar propulsion, such as effective flexibility (Wiggins & Goldstein 1998; Yu, Lauga & Hosoi 2006; Peng, Elfring & Pak 2017), swimmer geometry (Lauga 2007), flagellar curvature (Liu *et al.* 2020; Hu, Zhang & Shelley 2022) and hydrodynamic interactions (HIs) (Elfasi, Elimelech & Gat 2018; Hu & Meng 2024). On the other hand, studies that combine mechanical models of flagellar deformation with molecular motor dynamics focus on how beating waveforms are regulated (see the review by Gilpin, Bull & Prakash 2020).

Under dynamic swimming conditions, the modelling of microorganisms involves different complexities, especially in the coupling between the cell body and flagella. In addition to mechanical coupling at the flagellar base, flagella are also coupled to the cell body through long-range HIs. Therefore, combining flagellar beating with a moving cell body is crucial for understanding the swimming mechanisms. Simulations and experimental measurements on various microorganisms have reported a decrease in swimming speed with increasing body size, including spermatozoa (Higdon 1979), monotrichous bacteria (Kamdar *et al.* 2023; Liu, Chen & Xu 2025), and the ciliate *Volvox* (Omori, Ito & Ishikawa 2020). In particular, body–flagella HIs have been shown to influence both the swimming speed and efficiency of bacteria (Liu *et al.* 2025). For flagellated microalgae, we expect more complex and stronger body–flagella HIs due to significantly time-varying and closer spacing between them. Over a beating period, the cell body moves back and forth, entraining a considerable volume of fluid. At low Reynolds number (Re), this body-induced flow decays slowly in space and can significantly influence flagellar propulsion. Meanwhile, the flow induced by the flagella also modifies the drag force acting on the cell body. As a minimal swimming model, the three-sphere model, in which the flagellar beating is represented by spheres moving along closed orbits, provides insights into flagellar synchronisation (Friedrich & Jülicher 2012), run-and-tumble behaviour (Bennett & Golestanian 2013), and three-dimensional helical trajectories (Cortese & Wan 2021). Optimal gaits for swimming and feeding have been identified using numerical optimisation (Tam & Hosoi 2011), where the swimming dynamics is modelled via flow singularities placed along the flagella and at the centre of the cell body. Although these studies reveal different dynamic behaviours of microalgal swimming, the effect of body–flagella HIs on swimming performance remains elusive. In recent work (Hu & Meng 2024), we modelled the swimming dynamics of multiflagellated microalgae, and demonstrated that body–flagella HIs determine the hydrodynamic advantage of multiflagellarity. In particular, swimming efficiency increases monotonically with the number of flagella only when they are attached to the anterior side of the cell body. However, whether an optimal body size exists for microalgal swimming is still an open question.

In this work, we use numerical simulations to investigate the effect of body–flagella HIs on the swimming performance of *C. reinhardtii*. We prescribe the flagellar waveform based on experimental measurements, and incorporate both body–flagella and interflagellar HIs using a boundary element method. We find that body–flagella HIs can

Parameter	Symbol	Value
Beating frequency	f_0	49 Hz
Flagellum radius, length	r_0, L	0.125, 14 μm
Anchored angle	ϕ	0.05π
Semi-minor, semi-major axes	a, b	3.89, 5.95 μm

Table 1. Parameters of a representative *C. reinhardtii* cell measured in experiments. The value of flagellum radius is taken from Sager & Palade (1957).

significantly enhance the swimming performance and lead to an optimal body size for both speed and efficiency. Further analysis based on the three-sphere model reveals that body–flagella HIs are effectively non-reciprocal.

2. Simulation model

Our swimming model of *C. reinhardtii* consists of two flagella of length L , and a spheroidal cell body with semi-minor axis a and semi-major axis b . Planar waveforms of cells held stationary are measured experimentally via high-speed videography (figure 1a) and serve as inputs for numerical simulations. Physical parameters of a sample cell are given in table 1. We parametrise each flagellum by the arc length $s \in [0, L]$ along its centreline. From the measured waveform, we evaluate the velocities of the two flagella relative to the cell body, $\mathbf{U}_i(s, t)$ for $i = 1, 2$, at each recorded time instant. In this work, we focus on the normal breaststroke gait. Although a small phase lag (approximately 0.1π) exists between the *cis* and *trans* flagella (Wan, Leptos & Goldstein 2014), only a sufficiently large phase difference ($\gtrsim \pi/2$) can lead to a non-negligible rotation of the cell body (Geyer *et al.* 2013). Therefore, the two flagella are assumed to beat mirror-symmetrically, and the swimming model undergoes only translational motion. We further neglect the non-planar components of flagellar beating, as the resulting rotation frequency of the cell is much smaller than the flagellar beating frequency (Cortese & Wan 2021).

We simulate the dynamics of the model swimmer using a hybrid boundary element and regularised Stokeslet method (Smith 2009). The regularised Stokeslet is the exact solution to the Stokes equation subjected to a smoothed point force. With a specific smooth function, it can be expressed as (Cortez, Fauci & Medovikov 2005)

$$\mathbf{G}_\epsilon(\mathbf{x}_0, \mathbf{x}) = \frac{I(R^2 + 2\epsilon^2) + \mathbf{R}\mathbf{R}}{R_\epsilon^3}, \quad (2.1)$$

where ϵ is the regularisation parameter, $\mathbf{R} = \mathbf{x}_0 - \mathbf{x}$, and $R_\epsilon = \sqrt{R^2 + \epsilon^2}$.

We represent the motions of the cell body and the flagella by distributions of regularised Stokeslets on the body surface and along the flagellar centrelines, respectively. As verified by a test problem on a rigid scallop in Appendix A, this flagellar representation yields swimming dynamics that closely matches that obtained from a non-local slender body theory (Johnson 1980). Denote the body surface by S and the flagella centrelines by C_i with $i = 1, 2$. The disturbance velocity in the fluid domain generated by the motion of the cell body is given by

$$\mathbf{u}_b(\mathbf{x}_0) = \frac{1}{8\pi\mu} \int_S \mathbf{G}_\epsilon(\mathbf{x}_0, \mathbf{x}) \cdot \mathbf{f}_b(\mathbf{x}) \, \mathrm{d}\mathbf{x}, \quad (2.2)$$

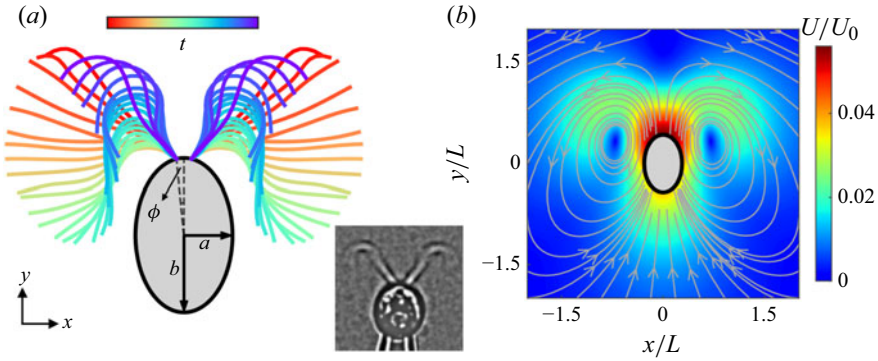


Figure 1. Model of swimming *C. reinhardtii*. (a) Flagellar waveform measured from experiments. Blue indicates recovery stroke, and red indicates power stroke. (b) Time-averaged disturbance flow field generated by the swimming model.

and that by the flagella is given by

$$\mathbf{u}_i(\mathbf{x}_0) = \frac{1}{8\pi\mu} \int_{C_i} \mathbf{G}_\epsilon[\mathbf{x}_0, \mathbf{x}_i(s)] \cdot \mathbf{f}_i(s) ds, \quad \text{for } i = 1, 2, \quad (2.3)$$

where μ is the fluid viscosity, $\mathbf{x}_i(s)$ is the centreline position, and $\mathbf{f}_b$ and $\mathbf{f}_i$ are the hydrodynamic force densities exerted by the body and flagella on the fluid, respectively. When the evaluation points $\mathbf{x}_0$ lie on the boundaries, the total disturbance velocity due to the flagella and cell body satisfies the following constraints imposed by the no-slip condition:

$$\mathbf{u}_b(\mathbf{x}_0) + \sum_i \mathbf{u}_i(\mathbf{x}_0) = \mathbf{U}_b(t), \quad \text{for } \mathbf{x}_0 \in S, \quad (2.4)$$

$$\mathbf{u}_b(\mathbf{x}_0) + \sum_j \mathbf{u}_j(\mathbf{x}_0) = \mathbf{U}_b(t) + \mathbf{U}_i(s, t), \quad \text{for } \mathbf{x}_0 \in C_i \quad \text{and} \quad i, j = 1, 2, \quad (2.5)$$

where $\mathbf{U}_b(t)$ is the instantaneous velocity of the cell body. Equations (2.4) and (2.5) incorporate both the interflagellar and the body–flagella HIs. Finally, the total hydrodynamic force and torque are zero:

$$\int_S \mathbf{f}_b(\mathbf{x}) d\mathbf{x} + \sum_i \int_{C_i} \mathbf{f}_i(s) ds = 0, \quad (2.6)$$

$$\int_S (\mathbf{x} - \mathbf{x}_b) \times \mathbf{f}_b(\mathbf{x}) d\mathbf{x} + \sum_i \int_{C_i} [\mathbf{x}_i(s) - \mathbf{x}_b] \times \mathbf{f}_i(s) ds = 0, \quad (2.7)$$

where the torques are evaluated relative to the body centre $\mathbf{x}_b$.

We scale lengths by L , time by the beating period T , velocities by $U_0 = L/T$, and forces by $F_0 = \mu L^2/T$. The body aspect ratio b/a is kept constant when varying the semi-major axis b . The integral formulations (2.2)–(2.7) are discretised into a system of linear equations for the unknowns $\mathbf{U}_b$, $\mathbf{f}_b$ and $\mathbf{f}_i$. Since the Stokeslet $\mathbf{G}_\epsilon(\mathbf{x}_0, \mathbf{x})$ often varies more rapidly near the source position $\mathbf{x}$ than the hydrodynamic forces, the evaluation of the Stokeslet is performed on a finer grid than that of the forces (Smith 2009). Equations (2.2) and (2.3) are approximated as

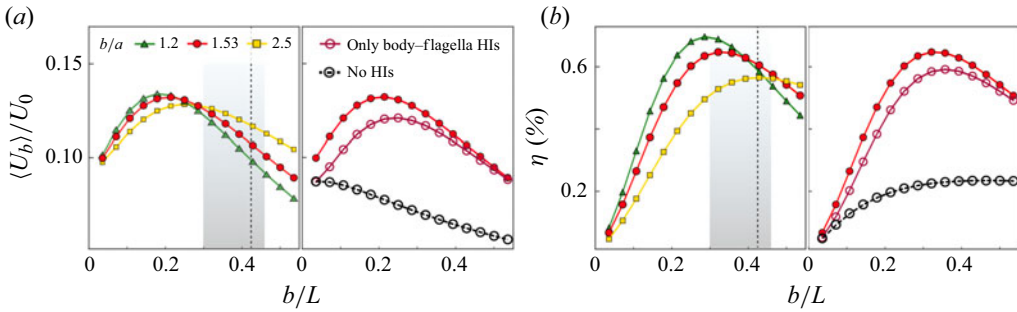


Figure 2. Effect of cell body size on swimming performance: (a) $\langle U_b \rangle / U_0$ as a function of b/L ; (b) swimming efficiency η as a function of b/L . The shaded areas indicate the distribution of b/L from experimental measurements. The vertical dashed lines mark the value of b/L of the sample cell. Other parameters are given in table 1.

$$\mathbf{u}(\mathbf{x}_0) = \frac{1}{8\pi\mu} \sum_{n=1}^N \mathbf{f}^n \cdot \int_{\delta S^n(\delta C^n)} \mathbf{G}_\epsilon(\mathbf{x}_0, \mathbf{x}) d\mathbf{x}, \quad (2.8)$$

where N is the number of discrete surface elements δS^n (line elements δC^n) on the cell body (on each flagellum) with force density $\mathbf{f}^n$. The integral of $\mathbf{G}_\epsilon$ over each element is evaluated using Gauss–Legendre quadrature with $m \times m$ points on δS^n (m points on δC^n). For most simulations, we use $N \approx 200$ on the sphere, $N = 25$ on each flagellum, and $m = 6$. The regularisation parameter ϵ is taken to be the same as the flagellum radius r_0 (table 1).

3. Results and discussion

3.1. Optimal cell body size for swimming

Figure 1(b) shows the disturbance velocity field of the model swimmer averaged over a beating period. There is one vortex on each side of the cell body, and a flow stagnation point exists in front of the swimmer along its swimming direction. These features closely resemble the measured flow field around free-swimming cells (Drescher *et al.* 2010). Using the parameters of the sample cell (table 1), we compute the time-averaged swimming speed $\langle U_b \rangle \approx 70.4 \mu\text{m s}^{-1}$, in agreement with the experimentally measured speed (Buchner *et al.* 2021). Slight variations in the anchored angle ϕ of the flagella (figure 1a) have little effect on the swimming performance.

We first evaluate the dependence of $\langle U_b \rangle$ on cell body size by varying the semi-major axis length b while using the same flagellar waveform data. As shown in figure 2(a), $\langle U_b \rangle$ varies non-monotonically with b/L , the ratio of body size to flagellar length. As the cell body becomes more slender (larger values of b/a), the maximum of $\langle U_b \rangle$ shifts to larger values of b/L . From a collection of approximately 40 cells, we measure an average $b/L = 0.38 \pm 0.08$, as indicated by the shaded regions in figure 2. For the sample cell ($b/a = 1.53$), the maximum occurs at $b/L \approx 0.21$, which is lower than its actual geometric ratio (0.425; see table 1) and lies more than one standard deviation below the measured average.

We further compute the swimming efficiency η using a definition similar to that in Lighthill (1952):

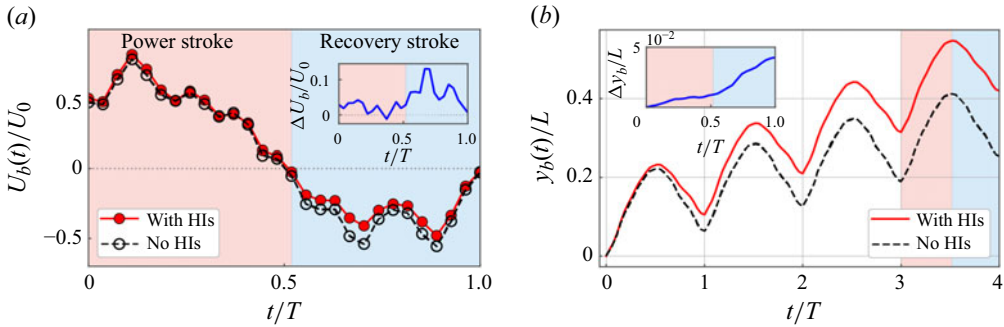


Figure 3. Effect of body-flagella HIs on (a) instantaneous swimming speed $U_b(t)$ and (b) instantaneous displacement $y_b(t)$. Insets show the differences between the results obtained with and without body-flagella HIs for (a) $U_b(t)$ and (b) $y_b(t)$ over a beating period. Red and blue regions indicate power and recovery strokes, respectively.

$$\eta = \frac{\zeta_b \langle U_b \rangle^2}{\left\langle \int_S \mathbf{f}_b \cdot \mathbf{U}_b \, d\mathbf{x} \right\rangle + \sum_i \left\langle \int_{C_i} \mathbf{f}_i \cdot (\mathbf{U}_b + \mathbf{U}_i) \, d\mathbf{s} \right\rangle}, \quad (3.1)$$

where ζ_b is the friction coefficient of a spheroid translating parallel to its long axis (Kim & Karrila 2013). Equation (3.1) expresses the ratio of the work required to drag a spheroid moving steadily at speed $\langle U_b \rangle$ to the work dissipated by the motions of the flagella and cell body. Figure 2(b) shows that η also exhibits a pronounced optimum with respect to b/L . Unlike $\langle U_b \rangle$, for the sample cell, the optimal value of b/L for η lies within one standard deviation of the measured average.

Removing the interflagellar HIs, which amounts to neglecting the disturbance velocity from the other flagellum in (2.5), has only a minor effect on the swimming performance (see the right-hand images in figure 2a,b). This is due to hydrodynamic screening by the cell body. Further removing the body-flagella HIs, i.e. neglecting $\mathbf{u}_i$ in (2.4) and $\mathbf{u}_b$ in (2.5), results in a much lower $\langle U_b \rangle$ and a monotonically decreasing trend with b . Therefore, the surprising increase of $\langle U_b \rangle$ at small values of b is attributed to increasingly strong body-flagella HIs as b increases. Meanwhile, the intrinsic viscous drag of the cell body $\zeta_b U_b$, also increases with b and eventually overwhelms the flagellar propulsion, leading to a decrease in $\langle U_b \rangle$ at sufficiently large b .

In figure 3, we compare the swimming speed $U_b(t)$ and the instantaneous displacement $y_b(t)$ of the cell body, obtained with and without body-flagella HIs. Our key observation is that body-flagella HIs not only increase the forward speed during the power stroke, but also reduce the backward speed during the recovery stroke (figure 3a); as a result, the swimmer achieves a larger forward displacement during the power stroke, and a smaller backward displacement during the recovery stroke (figure 3b). Over one period, the swimmer accumulates a net displacement of approximately $0.1L$, which is approximately twice that achieved without body-flagella HIs.

The appearance of an optimal body size for swimming is robust and not sensitive to variations in flagellar waveforms. In Appendix B, we present additional simulation results using waveforms measured from different sample cells. Despite large variations in the static curvature, wavelength and amplitude, the swimming performance consistently varies non-monotonically with b/L . This suggests that body-flagella HIs – reflecting how they influence each other's drag forces – are predominantly determined by their relative large-scale approaching and receding motions on the order of L during the power and recovery

strokes. The robustness also implies that our results are not likely to be affected by slight flagellar adaptations *in vivo* arising from flexibility and active forcing.

3.2. Three-sphere model of *Chlamydomonas* swimming

To reveal the mechanisms underlying the enhancement of swimming performance by body–flagella HIs, we consider a three-sphere model consisting of a central sphere of radius r_b , representing the cell body, and two side spheres of radius r_f , representing the flagella (figure 4a). The flagellar spheres are assumed to be much smaller than the cell body, $r_f \ll r_b$. Unlike previous works (Friedrich & Jülicher 2012; Bennett & Golestanian 2013), we explicitly account for the no-slip boundary condition on the body surface by using the spherical image system $\mathbf{G}_{im}$ (Kim & Karrila 2013). The two flagellar spheres, located at $\mathbf{x}_i$ ($i = 1, 2$), are prescribed to move with velocity $v_0 \hat{\mathbf{t}}_i$ relative to the body, and are mirror-symmetric with respect to the model centreline. Due to the force-free condition, the cell body, located at $\mathbf{x}_b$, develops an instantaneous translational velocity $\mathbf{U}_b$. The motion of the cell body induces a disturbance flow $\mathbf{v}_b$ in the surrounding fluid,

$$\mathbf{v}_b(\mathbf{x}_0, \mathbf{x}_b; \mathbf{U}_b, r_b) = \frac{3}{4} r_b \mathbf{U}_b \cdot \left(1 + \frac{r_b^2}{6} \nabla^2 \right) \mathbf{G}(\mathbf{x}_0, \mathbf{x}_b), \quad (3.2)$$

where $\mathbf{G}$ denotes the free-space Stokeslet.

The total drag force on the flagellar spheres is $\mathbf{F}_i = -6\pi\mu r_f(v_0 \hat{\mathbf{t}}_i + \mathbf{U}_b) + \mathbf{F}_{b \rightarrow i}$, where $\mathbf{F}_{b \rightarrow i}$ is the contribution to $\mathbf{F}_i$ by the cell body, i.e. the body-to-flagella (B-to-F) force, given by

$$\mathbf{F}_{b \rightarrow i} = 6\pi\mu r_f \left[\mathbf{v}_b(\mathbf{x}_i, \mathbf{x}_b; \mathbf{U}_b, r_b) + \frac{1}{8\pi\mu} \mathbf{G}_{im}(\mathbf{x}_i, \mathbf{x}_b; r_b) \cdot \mathbf{F}_i \right]. \quad (3.3)$$

In (3.3), $\mathbf{v}_b$ represents the dynamic component of the B-to-F force, and exists only when the cell body is in motion. The term involving $\mathbf{G}_{im}$ represents the static component of the B-to-F force, and captures the effect of a nearby no-slip boundary on the flagellar force. The total drag force on the cell body is $\mathbf{F}_b = -6\pi\mu r_b \mathbf{U}_b + \sum_i \mathbf{F}_{i \rightarrow b}$, where $\mathbf{F}_{i \rightarrow b}$ is the contribution to the drag force acting on the cell body by the flagella-induced flow, i.e. the flagella-to-body (F-to-B) force. Using the strength of the image Stokeslet in $\mathbf{G}_{im}$, $\mathbf{F}_{i \rightarrow b}$ can be expressed as (Higdon 1979; Kim & Karrila 2013)

$$\mathbf{F}_{i \rightarrow b} = - \left[\left(\frac{3r_b}{2R_i} - \frac{r_b^3}{2R_i^3} \right) \hat{\mathbf{R}}_i \hat{\mathbf{R}}_i + \left(\frac{3r_b}{4R_i} + \frac{r_b^3}{4R_i^3} \right) (\mathbf{I} - \hat{\mathbf{R}}_i \hat{\mathbf{R}}_i) \right] \cdot \mathbf{F}_i, \quad (3.4)$$

where $\mathbf{R}_i = \mathbf{x}_i - \mathbf{x}_b$ and $\hat{\mathbf{R}}_i = \mathbf{R}_i / |\mathbf{R}_i|$. The body velocity $\mathbf{U}_b$ can be obtained from the force-free condition $\mathbf{F}_b + \sum_i \mathbf{F}_i = 0$.

3.3. Effective non-reciprocal interactions between cell body and flagella

The three-sphere model (§ 3.2) enables us to separately evaluate the effects of the F-to-B force and the dynamic and static components of the B-to-F force. We consider the time instant when the three spheres form an isosceles right triangle. The left flagellar sphere moves in the direction $\hat{\mathbf{t}}_1 = (\cos \alpha, \sin \alpha, 0)$, where we set the angle relative to the x axis as $\alpha = 5\pi/4$. Reversing the direction of motion by setting $\alpha = \pi/4$ corresponds to switching from the power stroke to the recovery stroke, and reverses the sign of $\mathbf{U}_b$. Figure 4(b) shows that the F-to-B force alone can significantly reduce $\mathbf{U}_b$ compared with that without any HIs. This implies that the reduction in swimming speed during the recovery stroke

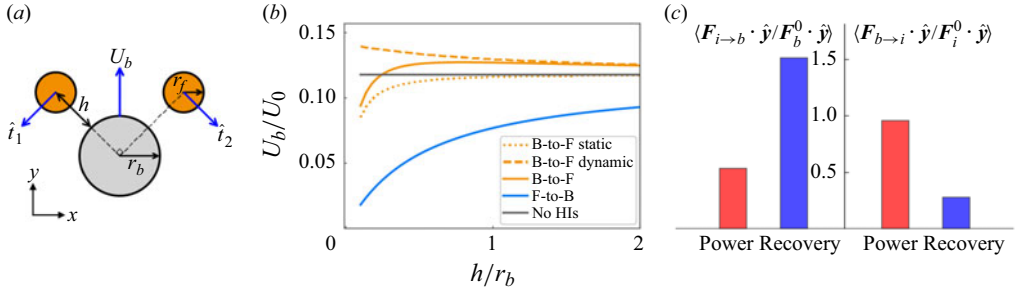


Figure 4. (a) Schematic of the three-sphere model. (b) Effect of different components of body–flagella HIs on the body speed U_b in the three-sphere model. The radius of the flagellar sphere is $r_f = 0.1 r_b$. (c) Average of the normalised drag forces $\langle \mathbf{F}_{i \rightarrow b} \cdot \hat{\mathbf{y}} / \mathbf{F}_b^0 \cdot \hat{\mathbf{y}} \rangle$ and $\langle \mathbf{F}_{b \rightarrow i} \cdot \hat{\mathbf{y}} / \mathbf{F}_i^0 \cdot \hat{\mathbf{y}} \rangle$ (see text) over the power and recovery strokes.

(figure 3) results from the impedance of the cell body by the flagella-induced flow. The dynamic component of the B-to-F force increases U_b , while the static component has the opposite effect but decays more rapidly with the distance from the body surface h . When $h \gtrsim r_b$, which is approximately the average separation between the flagella and cell body (figure 1a), the static component has a negligible effect on U_b . Therefore, the slight increase in U_b during the power stroke (figure 3a) may arise from the dominance of the dynamic component of the B-to-F force.

To verify the predictions from the three-sphere model, we evaluate how the cell body and flagella contribute to the drag forces on each other in the flagellated swimming model (§ 2). We approximate the spheroidal cell body as a sphere of radius $r_b = (a + b)/2$, and compute the F-to-B force $\mathbf{F}_{i \rightarrow b}(t)$ by integrating (3.4) along the arc length s , with $\mathbf{F}_i$ replaced by the flagellar force density $\mathbf{f}_i(s)$. The dynamic component of the B-to-F force is computed as

$$\mathbf{F}_{b \rightarrow i}(t) = \int_{C_i} [\zeta_{\parallel} \mathbf{p}_i \mathbf{p}_i + \zeta_{\perp} (\mathbf{I} - \mathbf{p}_i \mathbf{p}_i)] \cdot \mathbf{v}_b[\mathbf{x}_i(s, t), \mathbf{x}_b; U_b, r_b] ds, \quad (3.5)$$

where $\zeta_{\parallel}$ and $\zeta_{\perp}$ are the parallel and perpendicular drag coefficients of the slender flagella (Cox 1970), and $\mathbf{p}_i$ is the unit tangent vector. To compare the relative strength of the effects of $\mathbf{F}_{i \rightarrow b}$ on the cell body and $\mathbf{F}_{b \rightarrow i}$ on the flagella, we normalise $\mathbf{F}_{i \rightarrow b}$ by $\mathbf{F}_b^0(t) = -6\pi\mu r_b U_b(t)$, and $\mathbf{F}_{b \rightarrow i}$ by

$$\mathbf{F}_i^0(t) = - \int_{C_i} [\zeta_{\parallel} \mathbf{p}_i \mathbf{p}_i + \zeta_{\perp} (\mathbf{I} - \mathbf{p}_i \mathbf{p}_i)] \cdot [\mathbf{U}_b(t) + \mathbf{U}_i(s, t)] ds. \quad (3.6)$$

Here, $\mathbf{F}_b^0$ and $\mathbf{F}_i^0$ are the drag forces on the cell body and flagella, respectively, due to their own translational motions in a quiescent fluid. We normalise the components of $\mathbf{F}_{i \rightarrow b}$ and $\mathbf{F}_{b \rightarrow i}$ along the swimming direction $\hat{\mathbf{y}}$, and compute the average over the power and recovery strokes separately.

Figure 4(c) shows that the body–flagella HIs are effectively non-reciprocal during both the power and recovery strokes. During the power stroke, $\langle \mathbf{F}_{b \rightarrow i} \cdot \hat{\mathbf{y}} / \mathbf{F}_i^0 \cdot \hat{\mathbf{y}} \rangle$ is larger than $\langle \mathbf{F}_{i \rightarrow b} \cdot \hat{\mathbf{y}} / \mathbf{F}_b^0 \cdot \hat{\mathbf{y}} \rangle$, and the effect of the B-to-F force is stronger than that of the F-to-B force, leading to a slightly larger forward speed (figure 3a). During the recovery stroke, $\langle \mathbf{F}_{i \rightarrow b} \cdot \hat{\mathbf{y}} / \mathbf{F}_b^0 \cdot \hat{\mathbf{y}} \rangle$ is larger than $\langle \mathbf{F}_{b \rightarrow i} \cdot \hat{\mathbf{y}} / \mathbf{F}_i^0 \cdot \hat{\mathbf{y}} \rangle$, and the effect of the F-to-B force dominates, leading to a smaller backward speed. The reversal of non-reciprocity when switching from the power stroke to the recovery stroke (figure 4c) arises from the

time-non-reciprocal deformation of the flagella. During the power stroke, the flagella are extended straight and oriented perpendicular to the swimming direction, resulting in a stronger B-to-F effect. During the recovery stroke, the flagella are coiled, more aligned with the swimming direction, and positioned closer to the cell body, resulting in a stronger F-to-B effect.

4. Conclusion

We modelled the swimming dynamics of *C. reinhardtii* using numerical simulations based on a boundary element method and experimentally measured flagellar waveform. We find that the swimming performance is significantly enhanced by the body–flagella HIs. As the cell body size increases, an optimal body size for swimming appears due to the competition between enhanced body–flagella HIs and increased viscous drag on the cell body. The measured average body size of *C. reinhardtii* is close to the computed optimal value that maximises the swimming efficiency. Using the three-sphere model, we demonstrate that the body–flagella HIs are effectively non-reciprocal: the body affects the flagella more strongly than vice versa during the power stroke, while the reverse holds during the recovery stroke. As a result, the forward speed is larger than that without HIs during the power stroke, and the backward speed is smaller during the recovery stroke.

Our results provide a hydrodynamic interpretation for the characteristic body size of microalgae. The existence of an optimal cell body size also suggests that the relation between the swimming performance and cargo size may be non-monotonic for certain swimmers composed of multiple interacting components, such as the three-sphere swimmer, which moves through sequential changes in the linear separations between the spheres (Nasouri, Vilfan & Golestanian 2019). This optimality could inform laboratory designs of biohybrid microrobots (Zhang *et al.* 2024).

Non-reciprocity generally emerges in interactions mediated by velocity fields. Another example of non-reciprocal HIs is found in dense cilia arrays (Hickey, Golestanian & Vilfan 2023). Two other geometric features of eukaryotic microorganisms, namely the ratio of flagellar wave amplitude to wavelength and the body aspect ratio, also align with hydrodynamic predictions (Lisicki, Rodrigues & Lauga 2024). Finally, beyond hydrodynamic effects, various factors can shape cell morphology. In particular, biological constraints naturally impose a lower limit on body size, as essential organelles such as the nucleus and chloroplasts require a minimum volume to function.

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Declaration of interests. The authors report no conflict of interest.

Data availability statement. All the data supporting this work and the simulation codes that reproduce the dynamics of the rigid scallop model using both the hybrid BEM method and the non-local slender body theory are available at https://github.com/shiyuanhu/microalgal_swimming.

Appendix A

As a test problem for our implementation of the hybrid boundary element method (BEM) and regularised Stokeslet method, we consider a rigid scallop consisting of two slender

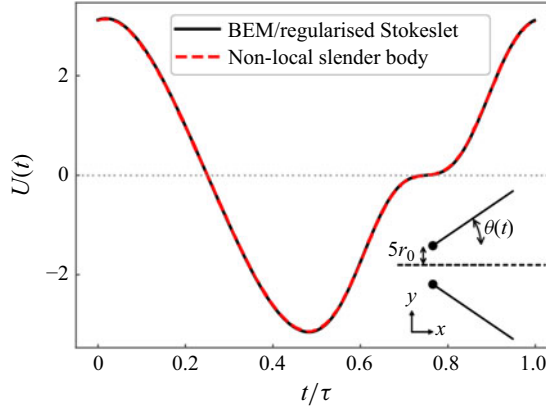


Figure 5. Comparison of the numerical results for a rigid scallop model, obtained from a non-local slender body theory and the hybrid BEM and regularised Stokeslet method used in this work. The two filaments are separated by a small distance at the hinge point to avoid overlapping. The number of line elements is $N = 100$, and $m = 6$.

rigid filaments hinged at a common end. This model does not swim on average, but oscillates. The orientation of the upper filament is prescribed as $\theta(t) = \theta_0[\sin(\omega t) + 1]$, where θ_0 is the amplitude, and ω is the frequency. The lower filament is mirror-symmetric.

From a non-local slender body theory (Johnson 1980), the velocity of the upper filament $\partial \mathbf{x}_1 / \partial t$ is related to the force densities $\mathbf{f}_i$ (for $i = 1, 2$) through

$$8\pi\mu \left[\frac{\partial \mathbf{x}_1(s, t)}{\partial t} - \mathbf{v}_{2 \rightarrow 1}(s, t) \right] = [c(\mathbf{I} + \mathbf{p}_1 \mathbf{p}_1) + 2(\mathbf{I} - \mathbf{p}_1 \mathbf{p}_1)] \cdot \mathbf{f}_1(s) + \mathbf{K}[\mathbf{f}_1](s), \quad (\text{A1})$$

where the slenderness parameter is $c = |\ln(\delta^2 e)|$, with δ the aspect ratio, the unit tangent vector is $\mathbf{p} = (\cos \theta, \sin \theta)$, and the integral operator $\mathbf{K}$ captures the non-local effects within the filament. The HIs between the two filaments are captured by

$$\mathbf{v}_{2 \rightarrow 1}(s) = \frac{1}{8\pi\mu} \int_0^L \frac{\mathbf{I} + \hat{\mathbf{R}}(s, s') \hat{\mathbf{R}}(s, s')}{R(s, s')} \cdot \mathbf{f}_2(s') ds', \quad (\text{A2})$$

where $\mathbf{R}(s, s') = \mathbf{x}_1(s) - \mathbf{x}_2(s')$. The filament velocity can be decoupled into a translation of the hinged point with velocity $U(t) \hat{\mathbf{x}}$ and a rotation with angular velocity $\dot{\theta}$.

We discretise the filament centreline using Chebyshev points. The operator $\mathbf{K}[\mathbf{f}](s)$ is first regularised (Tornberg & Shelley 2004) and then evaluated via Clenshaw–Curtis quadrature with spectral accuracy (Trefethen 2000). Equations (A1) and (A2) form a dense linear system for the unknowns $\mathbf{f}_i$ and $U(t)$. Figure 5 shows that the result obtained from the hybrid BEM method using a line distribution of regularised Stokeslets agrees closely with the non-local slender body theory.

Appendix B

We quantify the geometric properties of waveforms used in our simulations by separating the tangent angle $\theta(s, t)$ into static and dynamic components (Geyer *et al.* 2016). The static component is defined as the time average of the tangent angle over a beating period, $\bar{\theta}(s) = (1/T) \int_0^T \theta(s, t) dt$, and the dynamic component is defined as $\tilde{\theta}(s, t) = \theta(s, t) - \bar{\theta}(s)$. The static component $\bar{\theta}(s)$ is approximately linear in s , which corresponds to a circular shape.

Parameter	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Curvature C (rad μm^{-1})	-0.11	-0.12	-0.13	-0.17
Length L (μm)	13.97	10.82	13.03	9.42
Wavelength λ (μm)	13.76	11.82	13.88	11.30
Amplitude A (rad)	1.09	1.03	1.02	1.02
Frequency f (Hz)	49	45	43	47
Semi-minor a , semi-major axis b (μm)	3.89, 5.95	4.24, 4.61	3.91, 4.44	3, 3.53

Table 2. Variations in experimental waveforms explored in simulations. Dataset 1 is used in the main text.

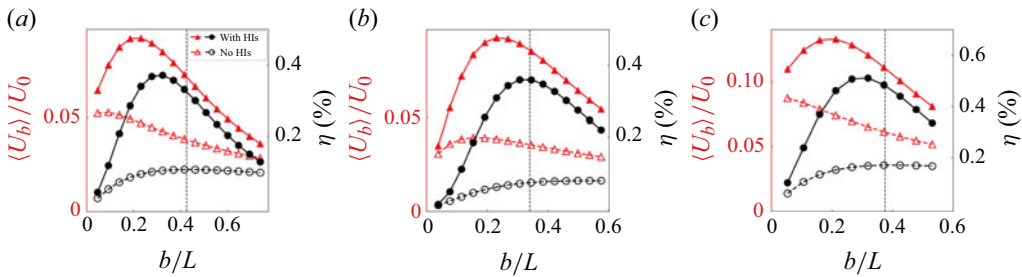


Figure 6. Effect of cell body size b/L on the swimming speed $\langle U_b \rangle / U_0$ and efficiency η , obtained from three additional waveforms measured in experiments (table 2): (a) dataset 2, (b) dataset 3, (c) dataset 4. The vertical dashed lines mark the values of b/L of each sample cell.

This static shape is characterised by its curvature $C = |\bar{\theta}(L) - \bar{\theta}(0)|/L$. The dynamic component $\tilde{\theta}(s, t)$ is close to a travelling wave with amplitude A and wavelength λ . These geometric parameters are summarised in table 2. Although the four waveforms differ significantly in curvature C , length L , and wavelength λ , the computed swimming speed $\langle U_b \rangle$ and efficiency η consistently display pronounced optima as a function of the body size b when HIs are incorporated (figure 6).

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